

Sequences

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Note: most of this worksheet is adapted from the 2022 version.

1 2018 Vietnam MO #1

The sequence (x_n) is defined as follows:

$$x_1 = 2, x_{n+1} = \sqrt{x_n + 8} - \sqrt{x_n + 3}$$

for all $n \geq 1$. Prove that

$$n \leq x_1 + x_2 + \cdots + x_n \leq n + 1.$$

Solution sketch: We see that the problem is easy if our initial condition were instead $x_1 = 1$; then, every term in the sequence is equal to 1. Motivated by this, we let $y_i = x_i - 1$, so that $y_1 = 1$ and $y_{n+1} = \sqrt{y_n + 9} - \sqrt{y_n + 4} - 1$. Here, we may show that the sequence y_n is *alternating*: that is, $y_n y_{n+1} < 0$ and $|y_{n+1}| < |y_n|$, after which the desired bound falls out; indeed, the right hand side becomes

$$y_n \left(\frac{1}{\sqrt{y_n + 9} + 3} - \frac{1}{\sqrt{y_n + 4} + 2} \right),$$

where the second term is around $\frac{1}{6} - \frac{1}{4} = -\frac{1}{12}$.

2 2022 ISL A1

Let $(a_n)_{n \geq 1}$ be a sequence of positive real numbers with the property that

$$(a_{n+1})^2 + a_n a_{n+2} \leq a_n + a_{n+2}$$

for all positive integers n . Show that $a_{2022} \leq 1$.

Solution sketch: We see that the terms a_n and a_{n+2} occur multiple times in the inequality. It would be nice if we could replace a_{n+1}^2 with 1, since that would make the inequality factor. So assume $a_{n+1} > 1$ for some n . Then, $a_n + a_{n+2} > a_n a_{n+2} + 1$, so $(a_n - 1)(a_{n+2} - 1) < 0$; that is, one of a_n and a_{n+2} is greater than 1, and the other is less than 1. Let's assume $a_n > 1$. Then, by the same logic, $a_{n-1} < 1$. Finally, the original equation provides some bound on a_n based on the value of a_{n+1} , so we want to shift it to produce a bound on a_{n+1} in terms of a_n . Ergo, we now add up the equations

$$(a_{n+1})^2 + a_n a_{n+2} \leq a_n + a_{n+2}$$

and

$$(a_n)^2 + a_{n-1} a_{n+1} \leq a_{n-1} + a_{n+1}$$

to get

$$a_n(a_n + a_{n+2} - 1) + a_{n+1}(a_{n-1} + a_{n+1} - 1) \leq a_{n-1} + a_{n+2},$$

which is a contradiction, since $a_n(a_n + a_{n+2} - 1) > a_{n+2}$ and $a_{n+1}(a_{n-1} + a_{n+1} - 1) > a_{n+1}$.

3 Recommended Problems

These problems are roughly in order of difficulty and equally distributed in difficulty relative to the later sections.

1. Let the sequence $\{a_n\}$ be defined recursively as follows:

$$a_1 = 1, a_2 = 1, a_{n+2} = a_{n+1} + \frac{1}{a_n}, n = 1, 2, \dots$$

Prove that $a_{180} > 19$.

2. Define the sequence of real numbers $x_1, x_2, \dots, x_{2021}$, such that x_1 is any real number and $x_n = 1 - x_1 x_2 \cdots x_{n-1}$ for all $n > 1$. Show that $x_{2021} > \frac{2021}{2022}$.
3. Suppose that a sequence $a_1, a_2, \dots$ of positive real numbers satisfies

$$a_{k+1} \geq \frac{k a_k}{a_k^2 + (k-1)}$$

for every positive integer k . Prove that $a_1 + a_2 + \dots + a_n \geq n$ for every $n \geq 2$.

4. Fix a positive integer k , and let $a_1, a_2, \dots, a_k$ be a sequence of nonnegative integers such that for any i, j with $i + j \leq k$, we have that $a_i + a_j \leq a_{i+j} \leq a_i + a_j + 1$. Classify all such sequences.
5. An increasing sequence of positive numbers $\dots, a_{-1}, a_0, a_1, \dots$ is given. For each positive integer k , let

$$b_k = \left\lceil \max_{i \in \mathbb{Z}} \frac{a_i + a_{i+1} + \dots + a_{i+k-1}}{a_{i+k-1}} \right\rceil.$$

Prove that either $b_k = k$ for all k , or the sequence b_k is eventually constant.

6. Let $a_0, a_1, \dots$ be an arbitrary infinite sequence of positive numbers. Show that the inequality $1 + a_n > \frac{n+1}{n} a_{n-1}$ holds for infinitely many positive integers n .
7. A sequence of real numbers $a_1, a_2, \dots$ satisfies the relation

$$a_n = - \max_{i_1 + i_2 + \dots + i_k = n} (a_{i_1} + a_{i_2} + \dots + a_{i_k}) \quad \text{for all } n > 2017.$$

Then, prove that this sequence is bounded.

8. Let $a_1, a_2, a_3, \dots$ be a sequence of positive real numbers, and s be a positive integer, such that

$$a_n = \max\{a_k + a_{n-k} \mid 1 \leq k \leq n-1\} \quad \text{for all } n > s.$$

Prove there exist positive integers $\ell \leq s$ and N , such that

$$a_n = a_\ell + a_{n-\ell} \quad \text{for all } n \geq N.$$

4 Easier Problems (IMO 0-1,4 level)

1. Let $a_0 < a_1 < a_2 \dots$ be an infinite sequence of positive integers. Prove that there exists a unique integer $n \geq 1$ such that

$$a_n < \frac{a_0 + a_1 + a_2 + \dots + a_n}{n} \leq a_{n+1}.$$

2. Let n be an positive integer. Find the smallest integer k with the following property; Given any real numbers $a_1, \dots, a_d$ such that $a_1 + a_2 + \dots + a_d = n$ and $0 \leq a_i \leq 1$ for $i = 1, 2, \dots, d$, it is possible to partition these numbers into k groups (some of which may be empty) such that the sum of the numbers in each group is at most 1.

3. Let $a_1, \dots, a_{2020}$ be a sequence of real numbers such that $a_1 = 2^{-2019}$, and $a_{n-1}^2 a_n = a_n - a_{n-1}$. Prove that $a_{2020} < \frac{1}{2^{2019}-1}$.
4. Let $a_1, a_2, \dots, a_n$ be a sequence of real numbers, and let m be a fixed positive integer less than n . We say an index k with $1 \leq k \leq n$ is good if there exists some ℓ with $1 \leq \ell \leq m$ such that $a_k + a_{k+1} + \dots + a_{k+\ell-1} \geq 0$, where the indices are taken modulo n . Let T be the set of all good indices. Prove that $\sum_{k \in T} a_k \geq 0$.
5. Let a_0 be an irrational number such that $0 < a_0 < \frac{1}{2}$. Define $a_n = \min\{2a_{n-1}, 1 - 2a_{n-1}\}$ for $n \geq 1$. Can it happen that $a_n > \frac{7}{40}$ for all n ? Let n be a positive integer, and consider a sequence $a_1, a_2, \dots, a_n$ of positive integers. Extend it periodically to an infinite sequence $a_1, a_2, \dots$ by defining $a_{n+i} = a_i$ for all $i \geq 1$. If

$$a_1 \leq a_2 \leq \dots \leq a_n \leq a_1 + n$$

and

$$a_{a_i} \leq n + i - 1 \quad \text{for } i = 1, 2, \dots, n,$$

prove that

$$a_1 + \dots + a_n \leq n^2.$$

5 Medium Problems (IMO 1,4-2,5 level)

1. Let $n \geq 3$ be a positive integer and let $(a_1, a_2, \dots, a_n)$ be a strictly increasing sequence of n positive real numbers with sum equal to 2. Let X be a subset of $\{1, 2, \dots, n\}$ such that the value of

$$\left| 1 - \sum_{i \in X} a_i \right|$$

is minimised. Prove that there exists a strictly increasing sequence of n positive real numbers $(b_1, b_2, \dots, b_n)$ with sum equal to 2 such that

$$\sum_{i \in X} b_i = 1.$$

2. For a sequence $x_1, x_2, \dots, x_n$ of real numbers, we define its *price* as

$$\max_{1 \leq i \leq n} |x_1 + \dots + x_i|.$$

Given n real numbers, Dave and George want to arrange them into a sequence with a low price. Diligent Dave checks all possible ways and finds the minimum possible price D . Greedy George, on the other hand, chooses x_1 such that $|x_1|$ is as small as possible; among the remaining numbers, he chooses x_2 such that $|x_1 + x_2|$ is as small as possible, and so on. Thus, in the i -th step he chooses x_i among the remaining numbers so as to minimise the value of $|x_1 + x_2 + \dots + x_i|$. In each step, if several numbers provide the same value, George chooses one at random. Finally he gets a sequence with price G .

Find the least possible constant c such that for every positive integer n , for every collection of n real numbers, and for every possible sequence that George might obtain, the resulting values satisfy the inequality $G \leq cD$.

3. We have 2^m sheets of paper, with the number 1 written on each of them. We perform the following operation. In every step we choose two distinct sheets; if the numbers on the two sheets are a and b , then we erase these numbers and write the number $a + b$ on both sheets. Prove that after $m2^{m-1}$ steps, the sum of the numbers on all the sheets is at least 4^m .
4. A sequence $x_1, x_2, \dots$ is defined by $x_1 = 1$ and $x_{2k} = -x_k, x_{2k-1} = (-1)^{k+1}x_k$ for all $k \geq 1$. Prove that $\forall n \geq 1, x_1 + x_2 + \dots + x_n \geq 0$.

5. Given an integer $n \geq 2$ and real numbers $x_1, x_2, \dots, x_n$ in the interval $[0, 1]$, prove that there exist real numbers $a_0, a_1, \dots, a_n$ satisfying the following conditions:

- $a_0 + a_n = 0$
- $|a_i| \leq 1$, for $i = 0, 1, \dots, n$
- $|a_i - a_{i-1}| = x_i$, for $i = 1, 2, \dots, n$.

6. Suppose that a sequence $\{a_n\}$ of integers has the following property: for all sufficiently large n , a_n equals the number of indices i , $1 \leq i < n$, such that $a_i + i \geq n$. Find the maximum possible number of integers which occur infinitely many times in this sequence.

7. Let $a_0, a_1, a_2, \dots$ be a sequence of real numbers such that $a_0 = 0, a_1 = 1$, and for every $n \geq 2$ there exists $1 \leq k \leq n$ satisfying

$$a_n = \frac{a_{n-1} + \dots + a_{n-k}}{k}.$$

Find the maximum possible value of $a_{2018} - a_{2017}$.

8. Find all positive integers n such that the following statement holds: Suppose real numbers $a_1, a_2, \dots, a_n, b_1, b_2, \dots, b_n$ satisfy $|a_k| + |b_k| = 1$ for all $k = 1, \dots, n$. Then there exists $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_n$, each of which is either -1 or 1 , such that

$$\left| \sum_{i=1}^n \varepsilon_i a_i \right| + \left| \sum_{i=1}^n \varepsilon_i b_i \right| \leq 1.$$

9. Prove that for $n > 1$ and real numbers $a_0, a_1, \dots, a_n, k$ with $a_1 = a_{n-1} = 0$,

$$|a_0| - |a_n| \leq \sum_{i=0}^{n-2} |a_i - ka_{i+1} - a_{i+2}|.$$

10. Let $a_1 < a_2 < a_3 < a_4 < \dots$ be an infinite sequence of real numbers in the interval $(0, 1)$. Show that there exists a number that occurs exactly once in the sequence

$$\frac{a_1}{1}, \frac{a_2}{2}, \frac{a_3}{3}, \frac{a_4}{4}, \dots$$

11. Find the smallest constant $C > 1$ such that the following statement holds: for every integer $n \geq 2$ and sequence of non-integer positive real numbers $a_1, a_2, \dots, a_n$ satisfying

$$\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n} = 1,$$

it's possible to choose positive integers b_i such that

- (i) for each $i = 1, 2, \dots, n$, either $b_i = \lfloor a_i \rfloor$ or $b_i = \lfloor a_i \rfloor + 1$, and
- (ii) we have

$$1 < \frac{1}{b_1} + \frac{1}{b_2} + \dots + \frac{1}{b_n} \leq C.$$

(Here $\lfloor \bullet \rfloor$ denotes the floor function, as usual.)

6 Hard Problems (IMO 2,5-3,6)

1. Let $n \geq 2$ be a positive integer and $a_1, a_2, \dots, a_n$ be real numbers such that

$$a_1 + a_2 + \dots + a_n = 0.$$

Define the set A by

$$A = \{(i, j) \mid 1 \leq i < j \leq n, |a_i - a_j| \geq 1\}$$

Prove that, if A is not empty, then

$$\sum_{(i,j) \in A} a_i a_j < 0.$$

2. The sequence $a_1, a_2, \dots$ of integers satisfies the conditions:

- $1 \leq a_j \leq 2015$ for all $j \geq 1$,
- $k + a_k \neq \ell + a_\ell$ for all $1 \leq k < \ell$.

Prove that there exist two positive integers b and N for which

$$\left| \sum_{j=m+1}^n (a_j - b) \right| \leq 1007^2$$

for all integers m and n such that $n > m \geq N$.

3. Let $f : \mathbb{N} \rightarrow \mathbb{N}$ be a function, and let f^m be f applied m times. Suppose that for every $n \in \mathbb{N}$ there exists a $k \in \mathbb{N}$ such that $f^{2k}(n) = n + k$, and let k_n be the smallest such k . Prove that the sequence $k_1, k_2, \dots$ is unbounded.
4. Let $x_0, x_1, \dots, x_{n_0-1}$ be integers, and let $d_1, d_2, \dots, d_k$ be positive integers with $n_0 = d_1 > d_2 > \dots > d_k$ and $\gcd(d_1, d_2, \dots, d_k) = 1$. For every integer $n \geq n_0$, define

$$x_n = \left\lfloor \frac{x_{n-d_1} + x_{n-d_2} + \dots + x_{n-d_k}}{k} \right\rfloor.$$

Show that the sequence $\{x_n\}$ is eventually constant.

7 Very Hard Problems (IMO 3-6+)

1. Suppose that $s_1, s_2, s_3, \dots$ is a strictly increasing sequence of positive integers such that the subsequences

$$s_{s_1}, s_{s_2}, s_{s_3}, \dots \quad \text{and} \quad s_{s_1+1}, s_{s_2+1}, s_{s_3+1}, \dots$$

are both arithmetic progressions. Prove that the sequence $s_1, s_2, s_3, \dots$ is itself an arithmetic progression.

2. For any finite sets X and Y of positive integers, denote by $f_X(k)$ the k^{th} smallest positive integer not in X , and let

$$X * Y = X \cup \{f_X(y) : y \in Y\}.$$

Let A be a set of $a > 0$ positive integers and let B be a set of $b > 0$ positive integers. Prove that if $A * B = B * A$, then

$$\underbrace{A * (A * \dots (A * (A * A)) \dots)}_{\text{A appears } b \text{ times}} = \underbrace{B * (B * \dots (B * (B * B)) \dots)}_{\text{B appears } a \text{ times}}.$$

3. For any two different real numbers x and y , we define $D(x, y)$ to be the unique integer d satisfying $2^d \leq |x - y| < 2^{d+1}$. Given a set of reals $\mathcal{F}$, and an element $x \in \mathcal{F}$, we say that the scales of x in $\mathcal{F}$ are the values of $D(x, y)$ for $y \in \mathcal{F}$ with $x \neq y$. Let k be a given positive integer. Suppose that each member x of $\mathcal{F}$ has at most k different scales in $\mathcal{F}$ (note that these scales may depend on x). What is the maximum possible size of $\mathcal{F}$?